Prodigy Anesthesia Quick Review

Bronchoscopy

When would the use of a rigid bronchoscope be preferable over a flexible/fiberoptic scope?	Rigid bronchoscopes are preferred over flexible scopes when the patient has massive hemoptysis, foreign bodies, vascular tumors, if the patient is a small child, or endobronchial resection is anticipated. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
List situations in which a flexible/fiberoptic scope is preferable over a rigid bronchoscope.	When a patient has pneumonia and selective cultures need to be obtained, for the positioning of double-lumen tubes, if the patient has limited mobility of the neck, for the use of difficult intubations, if bronchial blockade is needed, if endotracheal tube position needs to be checked, if the patient has peripheral and upper lobe lesions, or if the amount of hemoptysis is limited *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
For what disorders/symptoms must a patient be evaluated prior to the performance of a bronchoscopy?	Hemoptysis, coughing, chronic lung disease, bronchospasm, respiratory obstruction, and the inability to adequately clear secretions. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1056-1057.*
How is the anesthesia source connected to a rigid bronchoscope?	Through a side arm provided on the rigid scope. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
List complications associated with rigid bronchoscopy.	Hemorrhage, damage to the teeth, tracheal or bronchial perforation, bronchospasm, barotrauma, and subglottic edema. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
Is the incidence of complications higher with a rigid or fiberoptic bronchoscope?	Rigid bronchoscopy is associated with a higher incidence of complications. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
What are the disadvantages to the use of local anesthesia during the performance of a bronchoscopy?	Lack of cooperation by the patient, the poor tolerance of bleeding by an awake patient. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*

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What are the advantages to using local anesthesia for patients undergoing a bronchoscopy?	The patient remains awake and breathing spontaneously. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
How is apneic oxygenation performed during a rigid bronchoscope?	A small catheter is inserted just above the carina, and oxygen is insufflated at a rate of 10 to 15 L/min. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
What is the other use for the suction channel on the adult fiberoptic bronchoscope?	The suction channel can also be used to oxygenate and ventilate the patient. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
What is the average decline in PaO2 seen in patients who have undergone a fiberoptic bronchoscopy? How long is this decline seen?	A decline of 20 mm Hg is typically seen for 1 to 4 hours following the procedure. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
What is the disadvantage to the apnea and intermittent ventilation technique used during a rigid bronchoscopy?	The possibility of a leak around the brochoscope can lead to hypoventilation and an increase in CO2. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 567.*
What are the advantages of the Sanders Injection System of oxygenation during a rigid bronchoscope?	Since there is no requirement for an eyepiece on the rigid scope, continuous ventilation is possible. (During the apneic and intermittent ventilation technique, ventilation is only possible when the eyepiece is in place.) The duration of the bronchoscopy is reduced, but, the effectiveness of the technique allows the procedure to be extended if necessary. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
What are three disadvantages to using the Sanders Injection System during rigid bronchoscopy?	Variable FiO2 levels occur as a result of the entrainment of air from the oxygen jet. Assessment of adequate ventilation may be difficult. If the patient has poor lung compliance, inadequate ventilation may occur. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*

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Why should PEEP be discontinued prior to the passage of a fiberoptic bronchoscope?	The bronchoscope itself causes a PEEP effect, which could lead to barotrauma in ventilated patients in which PEEP is already applied. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
At what pressure is oxygen delivered with a Sanders Injection System?	50 psig *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*
What is the minimum endotracheal tube diameter in which an adult fiberoptic bronchoscope will pass?	7.0 mm *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1058.*
When apneic oxygenation is performed during a rigid bronchoscopy, how quickly does the PaCO2 increase?	The PaCO2 increases about 6 mmHg in the first minute and 3 mm Hg/min after that. CO2 accumulation during apneic oxygenation can lead to respiratory acidosis and cardiac dyshythmias. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1057.*

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